

**PRIME'S MINISTER OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
SONGEA DISTRICT COUNCIL
FORM FOUR PRE – NECTA EXAMINATION**

031/1

PHYSICS 1

Time: 3:00 Hours.

JULY 2026

INSTRUCTIONS

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1. This paper consists of sections **A, B** and **C** with a total of Eleven **(11) questions**.
2. Answer all questions in section **A** and **B** and two **(2) questions from section C**.
3. Section A carries Sixteen (16) marks, section B carries Fifty four (54) marks and section C carries Thirty **(30) marks**.
4. All writings must be in blue or black ink, except drawings which should be in pencil
5. Non - programmable calculators are allowed in the examination room.
6. Write your Examination Number on every page of your answer sheet (s).
7. Where necessary, the following constants may be used.
 - (i) Acceleration due to gravity, $g = 10 \text{ m/s}^2$
 - (ii) Density of water = 1.0 g/cm^3
 - (iii) π , = **3.14**.
 - (iv) Coefficient of linear expansivity of the brick = $1.2 \times 10^{-5} / \text{K}$
 - (v) Speed of light in air = $3 \times 10^8 \text{ m/s}$.
 - (vi) Speed of sound in air = **330 m/s**.
 - (vii) Density of mercury = 13.6 g/cm^3
 - (viii) Atmospheric pressure = $1.01 \times 10^5 \text{ Pa}$

Section A(16 Marks)

Answer **all** questions in this section.

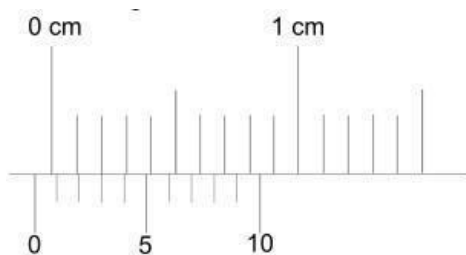
i. The scattered light may appear as a visible cone of faint, diffuse, and roughly triangular white glow in the sky called.

- A. Comets B. Constellation. C. Galaxies D. Zodiacal light.

ii. A liquid has a volume of 100cm^3 and a mass of 85 g. The density of water is 1.0g/cm^3 . How does the density of the liquid compare with the density of water?

- A. Its density is higher than that of water.
B. Its density is lower than that of water.
C. Its density is the same as that of water.
D. It is impossible to say with only this data.

iii. What is the zero error in figure below.



- A. 0.3 mm. B. 0.7 mm C. -0.3 mm D. -0.7 mm

iv. A force acting on an object causes some properties of the object to change. Which of the list contains properties that can be changed by the action of force?

- A. Mass, motion, and shape B. Mass, motion, and size
C. Mass, shape, and size D. Motion, shape and size

v. A sample of milk diluted with water has a density of 1040kg/m^3 . If pure milk has a density of 1080kg/m^3 , then the percentage of water by volume in the diluted milk is:

- A. 55% B. 40% C. 50% D. 45%

vi. A glass capillary tube is placed in a beaker of water. The water level inside the tube is higher than the water level outside due to :

A. Cohesion. B. Adhesion. C. Surface tension. D. Elasticity

vii. A cup like rubber sucker with a long handle at one end, used to unclog pipes and make them operate smoothly.

A. Lift pump B. Siphon C. Plunger D. Syringe

Viii. The deterioration caused by embedded impurities such as iron, lead and carbon due to the currents that are flowing from and to the same electrode (battery) is called?

A. Polarization B. Local action C. Depolarization. D. Dilute sulphuric acid.

ix. The following statements about magnetic lines of force are correct EXCEPT.

A. Always form close loops. B. Start at North pole and end at the south pole.

C. Cross one another. D. Stronger when the lines are closer together.

X. The displacement of an object for a round trip between two locations.

A. Is always greater than zero B. Is always less than zero C. Is zero. D. Can have any value.

2. Match the descriptions of the terms used in Application of vectors in List A with the corresponding concepts used in applications of vectors in List B by writing the letter of the correct response beside the item number in the answer sheet (s) provided. **(6 marks)**

LIST A	LIST B
(i) Vectors that act on the same plane	A. Resultant vector
(ii) A single vector that is a net effect of two or more vectors of the same kind.	B. Equilibrant vector
(iii) A set of coordinates used to calculate the position and velocity of an object.	C. Relative Velocity
(iv) A force that has the same magnitude as the resultant vector but acts in opposite direction.	D. Resolution of vector
(v) A process of splitting or separating a single vector into its components which are usually perpendicular to each other.	E. Absolute velocity
(vi) A movement of an object as seen by an observer on another frame of reference.	F. Relative motion
	G. Frame of reference
	H. Coplanar vectors
	I. Orthogonal vectors

SECTION B (54 Marks)

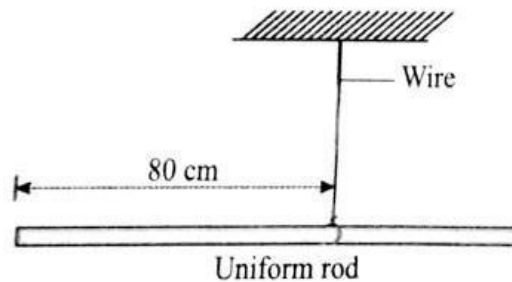
Answer **all** questions in this section.

3. (a) Is there any change in the direction of the ray of light as it emerges from the other side of the glass block? Explain. **(4 marks)**

(b). A compound microscope consists of the objective lens and the eyepiece lens of focal length 12cm and 4cm, respectively. The two lenses are separated by a distance of 30cm. The microscope is focused so that the image is formed at infinity. Determine the position of the object. **(5 marks)**

4.(a) Why is it not be practical to make a barometer with water instead of mercury? **(4 marks)**

(b) A uniform rod with a mass 120 g and a length of 130 cm is suspended by a wire from a point 80 cm from one end as shown in. **figure2.0**

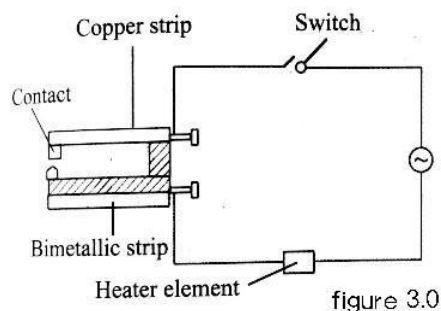


What mass must be hung on the rod for it to be in equilibrium? What will be the tension (**T**) in the wire? **(5 marks)**

5. (a) You observe a stone and leaf place in a river. One sinks rapidly , and the other gently floats on the water's surface. Explain the factors responsible for this difference in behavior. **(4 marks)**

(b). The input force of 11 N acting on the effort arm of a lever moves 0.4 m, which lifts a 40 N weight resting on the resistance arm a distance of 0.1 m. What is the efficiency of the machine? **(5marks)**

6.(a) **Figure 3.0** shows a circuit diagram for controlling the temperature of a room.



- (i). Explain the function of the bimetallic strip. **(2marks)**
- (ii). Describe how the circuit controls the temperature when the switch is closed.**(2marks)**
- (b). When cooking using firewood, the smoke produced do not always rise upwards. What is the reason for this? **(5 marks)**
7. (a) Sound of explosions taking place on other planets is not heard by a person on the Earth. Explain. **(4marks)**
- (b) The element tin has a total of twenty five isotopes with the lightest isotope being $^{108}_{50}\text{Sn}$. Assuming that all twenty five isotopes exist and the difference in mass numbers of two successive isotopes is one, write down the symbol for the heaviest isotope. **(5marks)**
8. (a) A seismometer located in Dodoma recorded an earthquake which originated in Mbeya. Which type of wave was recorded first? Give reasons for your choice. **(3 marks)**
- (b) Describe the three uses of constellations.**(6 marks)**

SECTION C (30 Marks)

Answer **two (2)** questions from this section.

9. (a) A palm of hand is held close the plate of a charged gold-leaf electroscope and slowly . Moved across it. State and explain what happen. **(05 marks)**
- (b) A cell of 6V emf and negligible internal resistance is connected to a resistor and drive a current of 3A through it. A cell of emf 1.5V is inserted in the circuit in series with the first one. The current remains at 3A .What is the internal resistance of the second cell? **(05 marks)**
- (c) A 240v washing machine has a motor rated at 250W and a heating element at 3KW. Find the annual cost for electricity of running it for 1.5hours per week assuming that, during this time the motor is on all time and the heater element only two- thirds of the time (Electricity cost 5Tsh per Kwh) **(05marks)**

10.(a) Describe the essential features of cathode ray tube using thermionic emission.(02 marks)
What the effect on the beam of the particle

(i) Of having a hotter filament (01 marks)

(ii) Of increasing the anode voltage (01 marks)

(b) The half life of a radioactive element is 10 minute. Calculate how it takes for 90% of a given mass of the element to decay.(04 marks)

(c) The P- type semiconductors are made from pure semiconductor by process known as “Doping”

(i) What is meant by doping? (01marks)

(ii) Explain how doping produce an n- type semiconductor (04 marks)

(iii) Name two materials used for doping (02 marks)

11.(a) Consider two organ pipes A and B sounded with their ,

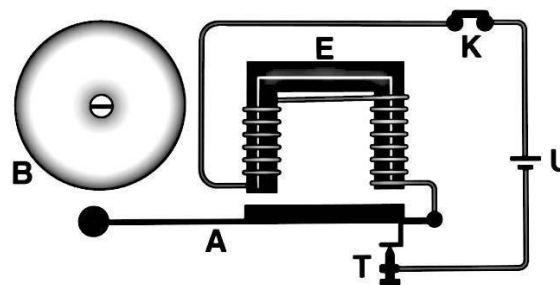
(i) Fundamental notes and

(ii) First overtones.

If a pipe A is closed at one end and pipe B is open at both ends, draw the diagrams to show the mode of vibration in each of the mode of the four cases. (05 marks)

(b). Show how Lenz's law /s a special case of the law of conservation of energy.(03 marks)

(c). Figure 04 illustrates an electric bell operated by a battery.



i) Name the parts labeled A, B, E, K,T and U. (03 marks)

(ii)State and explain what happens to the "A “when the switch is closed.(04 marks)